

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- | | | |
|-------------------------------------|-------------------------------------|--|
| ☐ | ☒ | The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ | The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☐ | ☒ | A description of all covariates tested |
| ☐ | ☒ | A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ | For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ | Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection R (v.4.4.1).

Data analysis R (v.4.4.1).
ropls v.1.36.0
glmnet v.4.1-8
caret v.6.0.94
pROC v.1.18.5
mediation 4.5.0
The code is available at https://github.com/sarastinson/olink_holbaek.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All results from statistical and bioinformatics analysis are provided in Supplementary Tables 1–11. The mean levels of proteins are available in the GitHub repository at https://github.com/sarastinson/olink_holbaek/tree/main/median_data. In accordance with the General Data Protection Regulation (GDPR, <https://gdpr-info.eu/>), individual-level clinical and proteomics data generated in this study cannot be made publicly available to maintain patient confidentiality. However, proteomics datasets can be requested from the authors by contacting T.H. at torben.hansen@sund.ku.dk. The obesity management protocol can be requested from J.-C.H. at jhom@regionsjaelland.dk. Access to the data can be granted through the Danish Data Protection Agency and the ethics committee for the Region Zealand, Denmark, subject to obtaining proper approvals and adherence to patient information and data processing agreements. The response time for data requests is within 1 month. Accessing and processing the data requires compliance with the following conditions: (1) a data-processing agreement must be signed between the data controller and processor; (2) data may only be processed for statistical and scientific studies; (3) personal data must be deleted, anonymized and destroyed at the end of the investigation; and (4) data must not be shared with unauthorized third parties or individuals.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

We used sex for biological definition of girls and boys. We include sex as a covariate in all analyses, as well as explored sex-specific differences in protein and cardiometabolic risk factor associations, through interaction models.

Reporting on race, ethnicity, or other socially relevant groupings

The study was conducted in a majority Danish European pediatric population, with ethnicity determined by self-report.

Population characteristics

In the cross-sectional study, 4,024 children and adolescents with baseline examinations and available blood samples (overweight/obesity, $n = 2,377$, 54.1% girls, median age = 11.7 years [9.6, 14.1]; and normal weight individuals from the population-based cohort, $n = 1,647$, 55.5% girls, median age = 11.5 years [9.0, 14.3]) were included. In the intervention study, 184 participants (54% girls, median age = 11.6 years [QR: 9.9–13.5]), who underwent obesity management and had a median follow-up period of 1.1 years, with an available blood sample at both time points were included.

Recruitment

The study population was derived from The HOLBAEK Study. The cross-sectional study is composed of two cohorts: 1) an obesity clinic cohort, including children and adolescents with a BMI ≥ 90 th percentile (BMI SDS ≥ 1.28), who participated in a comprehensive, multidisciplinary non-pharmacological obesity management program at the Children's Obesity Clinic, Holbaek Hospital; and 2) a population-based cohort, recruited from the school system across 11 municipalities in Zealand, Denmark. For the intervention study, children and adolescents with overweight or obesity (BMI SDS ≥ 1.28) enrolled in the multidisciplinary, family-based and individual-centered obesity clinic cohort received comprehensive management using an evidence-based treatment protocol.

This study may suffer from selection bias. Families attending the obesity clinic may have higher awareness of obesity and those that consented to participate may differ in socioeconomic background and other demographic factors than those who declined to participate. While the population cohort was recruited from the school system, it was limited to the Zealand region of Denmark, which may affect the generalization of the findings.

Ethics oversight

According to the Declaration of Helsinki, written informed consent was obtained from all participants. An informed oral assent was given by the participant if the participant was younger than 18 and then the parents gave informed written consent. The study was approved by the Ethics Committee of Region Zealand, Denmark (SJ-104) and by the Danish Data Protection Agency (REG-043-2013). The HOLBAEK Study is registered at ClinicalTrials.gov (NCT00928473).

We obtained access to UK Biobank data under accession number 32683, which has received ethical approval from the National Health Service National Research Ethics Service (ref 11/NW/0382), with informed consent obtained from all participants. Proteomic profiling of the UK Biobank was approved by the Access Subcommittee of UK Biobank, under Access Management System Application No. 65851.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

- ☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The cross-sectional study included 2,377 children and adolescents with overweight/obesity and 1,647 with normal weight, which was determined as all baseline individuals with a blood sample and genotypes available. The intervention study included 200 children and adolescents with overweight or obesity, who were followed for a median of 1.1 years (IQR 1.0–1.2) and 184 had available blood samples at follow-up for proteomics analysis. No sample size calculation was performed. To our knowledge this is the largest proteomics study performed in pediatric population to date. The sample size of the intervention study was restricted by feasibility, including clinical time and the budget available for proteomics analysis.
Data exclusions	Among 4,133 children and adolescents with available plasma proteomics in the cross-sectional study, participants were excluded based on diagnosed type 1 diabetes mellitus or T2D (n=7); intake of medications including statins, insulin, metformin and liraglutide (n=8); meeting T2D criteria based on blood sample (fasting plasma glucose ≥ 7.0 mmol l ⁻¹ and/or HbA1c ≥ 48 mmol mol ⁻¹) (n=8); the interval between visit and blood sample collection >90 days (n=30); underweight (BMI <5 th percentile [BMI SDS <-1.64]) (n=53). As a result, 4024 participants were included in the cross-sectional analysis.
Replication	We performed external replication in an independent cohort, using the UK Biobank, where Olink Explore 3072 PEA data was available. The direction of effects was largely consistent across the two cohorts. Notably, some proteins displayed significant associations in The HOLBAEK Study but not in the UK Biobank.
Randomization	Complete randomization across batches was performed, considering cohort as a factor for the proteomics analysis. Other clinical tests did not require randomization, as data collection happened at the same time as the clinical visit. We do not present a randomized clinical trial in the current study.
Blinding	Researchers performing the proteomics analysis were blinded with no access to the clinical data of individuals. For experiments other than the proteomics, i.e. clinical recruitment and sample collection, blinding was not relevant, as this includes a cross-sectional study and an intervention study, where all participants received the same treatment.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	A full list of proteins measured using the antibody-based Olink Target 96 (Cardiovascular II and Inflammation panels) and Explore 3072 is provided on the Olink website: https://olink.com/products/olink-target-96 and https://olink.com/products/olink-explore-3072-384 . All assays in Olink's panels use antigen affinity-purified polyclonal or monoclonal antibodies (or combinations of both), with the majority being commercially available.
Validation	Validation data for the Target 96 and Explore 3072 panels are available on the Olink website: https://olink.com/products/olink-target-96 and https://olink.com/products/olink-explore-3072-384 .

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration	NCT00928473
Study protocol	The study protocol was approved by the Ethics Committee of Region Zealand, Denmark (SJ-104). It is available upon request from J.-C.H. at jhom@regionsjaelland.dk .
Data collection	Participants from The HOLBAEK Study were enrolled between January 2009 and April 2019. Clinical data was collected at the

participant's baseline visit and 1-year follow-up visit, at The Children's Obesity Clinic Department of Pediatrics, Holbaek Hospital, Denmark.

Outcomes

In the cross-sectional study, the main outcome was identifying circulating proteins associated with BMI SDS and cardiometabolic risk factors. In the intervention study, the primary outcome was identifying associations between changes in circulating proteins and reductions in BMI SDS, as well as changes in cardiometabolic risk profiles.

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.